

Lecture 3: A first look at dimension reduction

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Dimension reduction and Predictive Modeling

One version of the curse of dimensionality

Samples tend to be further away from the origin

Let $\mathbf{x} \in [-1, 1]^p$ be a uniformly distributed random variable. For $0 \leq t \leq 1$ consider

$$q = \mathbb{P}(-t \leq x^{(1)} \leq t, \dots, -t \leq x^{(p)} \leq t) = \left(\frac{2t}{2}\right)^p$$
$$\Rightarrow t = q^{1/p}$$

In a large enough sample about q percent of observations will be in $[-t, t]^p$.

In high dimensions, most data points are far away from the origin.

How should t be chosen so that about q percent of observations lie in $[-t, t]^p$?

p	$q = 1\%$	$q = 10\%$
2	$t \approx 0.01$	$t \approx 0.32$
3	$t \approx 0.22$	$t \approx 0.46$
10	$t \approx 0.63$	$t \approx 0.79$
100	$t \approx 0.95$	$t \approx 0.98$

Another version of the curse of dimensionality

Pairwise distances grow with dimension

If $\mathbf{x}, \mathbf{y} \in [0, 1]^p$ uniformly distributed, then their pairwise distance $\|\mathbf{x} - \mathbf{y}\|_2$ grow with p .

The last column suggests that the mean pairwise distance grows as $O(\sqrt{p})$.

The standard deviations stay constant suggesting that observations have increasingly similar pairwise distances in high dimensions.

Mean and standard deviation of the pairwise distances of $n = 500$ simulated observations.

p	Mean	SD	Mean / $\sqrt{p}$
2	0.52	0.25	0.37
3	0.66	0.25	0.38
10	1.28	0.25	0.40
100	4.07	0.24	0.41
500	9.13	0.25	0.41
1000	12.91	0.24	0.41

High-dimensional predictive modeling

What does a predictive model do? The methods we have discussed so far try to capture some type of local, average behaviour - meaning, observations that are close borrow information from each other to come to a decision regarding the value of the outcome variable (numerical or class label). [.5em]

In high-dimensional settings, the notion of neighborhood breaks down.

In addition, many of the methods we have discussed will either be numerically unstable or ill-defined (i.e., can't estimate parameters) if the data dimension is too large.

What can be done about this dilemma?

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1. **Feature selection:** Deciding on a subset of the original features
 - ▶ We will talk a lot about feature selection later in class
 - ▶ Simplest strategy is some kind of pre-processing or *filtering*
 - ▶ Example: max variance features, features that are statistically associated with the label (t-test, ANOVA, correlation)

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1. **Feature selection:** Deciding on a subset of the original features
 - ▶ We will talk a lot about feature selection later in class
 - ▶ Simplest strategy is some kind of pre-processing or *filtering*
 - ▶ Example: max variance features, features that are statistically associated with the label (t-test, ANOVA, correlation)
2. **Feature transformation:** Combining existing features while reducing dimension (e.g. PCA)
 - ▶ The feature transformation might destroy/obscure relationships in the original data that it cannot capture
 - ▶ Since features are transformed, it is not guaranteed that uninformative features are actually filtered out

High-dimensional predictive modeling

When p is large

- ▶ kNN: notion of neighborhood breaks down

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- ▶ logistic regression: ill-defined/multicollinearity or numerically unstable model
- ▶ discriminant analysis: ill-defined/multicollinearity or numerically unstable model

Principal Component Analysis

Projection onto a subspace

Assume $\mathbf{x} \in \mathbb{R}^p$. Given **orthonormal vectors** $\mathbf{b}_1, \dots, \mathbf{b}_m$, i.e.

$$\|\mathbf{b}_j\| = 1 \quad \text{and} \quad \mathbf{b}_j^\top \mathbf{b}_k = 0 \text{ for } j \neq k$$

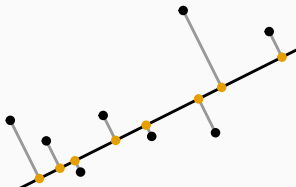
where $m < p$, the projection of $\mathbf{x}$ onto the m -dimensional linear subspace $V_m = \text{span}(\mathbf{b}_1, \dots, \mathbf{b}_m)$ is

$$\hat{\mathbf{x}} = \sum_{j=1}^m (\mathbf{x}^\top \mathbf{b}_j) \mathbf{b}_j = \underbrace{\left(\sum_{j=1}^m \mathbf{b}_j \mathbf{b}_j^\top \right)}_{\text{Projection matrix}} \mathbf{x}$$

The projection is **orthogonal**, i.e.

$$(\mathbf{x} - \hat{\mathbf{x}})^\top \mathbf{b}_j = 0$$

for all $\mathbf{b}_j$.



Rayleigh Quotient

Let $\mathbf{A} \in \mathbb{R}^{k \times k}$ be a symmetric matrix. For $\mathbf{0} \neq \mathbf{x} \in \mathbb{R}^k$ define

$$J(\mathbf{x}) = \frac{\mathbf{x}^\top \mathbf{A} \mathbf{x}}{\mathbf{x}^\top \mathbf{x}}$$

$J(\mathbf{x})$ is called the **Rayleigh Quotient** for $\mathbf{A}$.

Maximizing the Rayleigh Quotient

The maximization problem

$$\max_{\mathbf{x}} J(\mathbf{x}) \quad \text{subject to} \quad \mathbf{x}^\top \mathbf{x} = 1$$

is solved by a **unit eigenvector** $\mathbf{x}$ of $\mathbf{A}$ corresponding to the **largest eigenvalue** λ of $\mathbf{A}$.

Note: $-\mathbf{x}$ is also a solution.

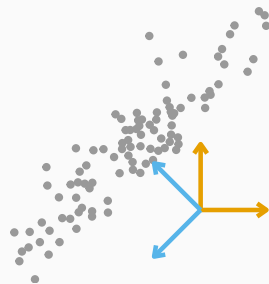
Principal Component Analysis (PCA) (I)

Goal: Given continuous data, find an orthogonal coordinate system such that the variance of the data is maximal along each direction.

Given data points $\mathbf{x}_1, \dots, \mathbf{x}_n$ and a unit vector $\mathbf{r}$, the **variance of the data along $\mathbf{r}$** is

$$S(\mathbf{r}) = \sum_{l=1}^n (\mathbf{r}^\top (\mathbf{x}_l - \bar{\mathbf{x}}))^2 = (n-1) \mathbf{r}^\top \hat{\Sigma} \mathbf{r}$$

where $\hat{\Sigma}$ is the empirical covariance matrix.



Axes

➡ Cartesian ➡ Principal Component

Principal Component Analysis (PCA) (II)

Direction with maximal variance: Find $\mathbf{r}$ such that

$$\max_{\mathbf{r}} S(\mathbf{r}) \quad \text{subject to} \quad \|\mathbf{r}\|^2 = \mathbf{r}^T \mathbf{r} = 1$$

- ▶ This is the same problem as maximizing the **Rayleigh Quotient** for the matrix $\hat{\Sigma}$.
- ▶ The **solution** is the eigenvector $\mathbf{r}_1$ of $\hat{\Sigma}$ corresponding to the largest eigenvalue λ_1 .

How do we find the other directions?

Project data on orthogonal complement of $\mathbf{r}_1$, i.e.

$$\hat{\mathbf{x}}_l = (\mathbf{I}_p - \mathbf{r}_1 \mathbf{r}_1^T) \mathbf{x}_l$$

and repeat the procedure above.

Intermezzo: Pre-processing

Data is often pre-processed before it is used in computational methods.

Given a data matrix $\mathbf{X} \in \mathbb{R}^{n \times p}$, let

- ▶ $\mathbf{m}_r \in \mathbb{R}^n$ be the vector of row-means,
- ▶ $\mathbf{m}_c \in \mathbb{R}^p$ be the vector of column-means, and
- ▶ $\mathbf{s} \in \mathbb{R}^p$ be the vector of per-column standard deviations.

Then (with $\mathbf{1}_n = (1, \dots, 1)^\top \in \mathbb{R}^n$)

- ▶ the matrix $\mathbf{X} - \mathbf{m}_r \mathbf{1}_p^\top$ has row means zero (**row-centred**),
- ▶ the matrix $\mathbf{X} - \mathbf{1}_n \mathbf{m}_c^\top$ has column means zero (**column-centred**), and
- ▶ the matrix $\mathbf{X} \text{diag}(1/\mathbf{s})$ has column standard deviations one (**standardised columns**)

Principal Component Analysis (PCA) (III)

Computational Procedure:

1. **Centre** (and possibly **standardise**) the columns of the data matrix $\mathbf{X} \in \mathbb{R}^{n \times p}$
2. Calculate the **empirical covariance matrix** $\hat{\Sigma} = \frac{1}{n-1} \mathbf{X}^T \mathbf{X}$
3. Determine the **eigenvalues** λ_j and corresponding orthonormal **eigenvectors** $\mathbf{r}_j$ of $\hat{\Sigma}$ for $j = 1, \dots, p$ and order them such that

$$\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_p \geq 0$$

4. The vectors $\mathbf{r}_j$ give the direction of the **principal components (PC)** $\mathbf{r}_j^T \mathbf{x}$ and the eigenvalues λ_j are the **variances along the PC directions**

Note: Set $\mathbf{R} = (\mathbf{r}_1, \dots, \mathbf{r}_p)$ and $\mathbf{D} = \text{diag}(\lambda_1, \dots, \lambda_p)$ then

$$\hat{\Sigma} = \mathbf{R} \mathbf{D} \mathbf{R}^T \quad \text{and} \quad \mathbf{R}^T \mathbf{R} = \mathbf{R} \mathbf{R}^T = \mathbf{I}_p$$

PCA and Dimension Reduction

Recall: For a matrix $\mathbf{A} \in \mathbb{R}^{k \times k}$ with eigenvalues $\lambda_1, \dots, \lambda_k$ it holds that

$$\text{tr}(\mathbf{A}) = \sum_{j=1}^k \lambda_j$$

For the empirical covariance matrix $\hat{\Sigma}$ and the variance of the j -th feature $\text{Var}[x_j]$

$$\text{tr}(\hat{\Sigma}) = \sum_{j=1}^p \text{Var}[x_j] = \sum_{j=1}^p \lambda_j$$

is called the **total variation**.

Using only the first $m < p$ principal components leads to

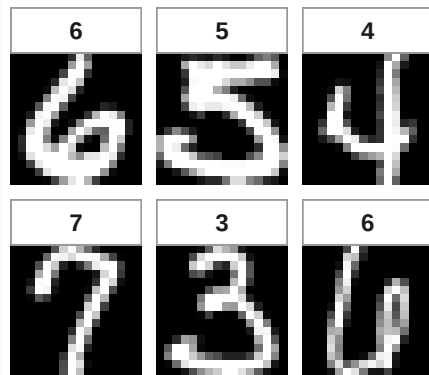
$$\frac{\lambda_1 + \dots + \lambda_m}{\lambda_1 + \dots + \lambda_p} \cdot 100\% \quad \text{of } \mathbf{explained\ variance}$$

PCA and Dimension Reduction: Example (I)

Variant of the MNIST handwritten digits dataset

($n = 7291$, 16×16 greyscale images, i.e. $p = 256$)

Digit	Frequency
0	0.16
1	0.14
2	0.10
3	0.09
4	0.09
5	0.08
6	0.09
7	0.09
8	0.07
9	0.09



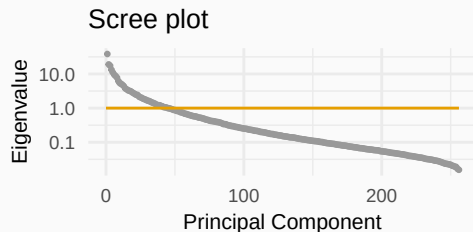
PCA and Dimension Reduction: Example (II)

For standardized variables

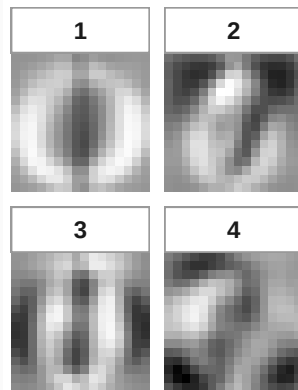
$$\text{tr}(\hat{\Sigma}) = p$$

Typical selection rule: Components with

$$\lambda_j \geq \frac{1}{p} \text{tr}(\hat{\Sigma}) \quad (= 1)$$



Visualisations of the first four principal components

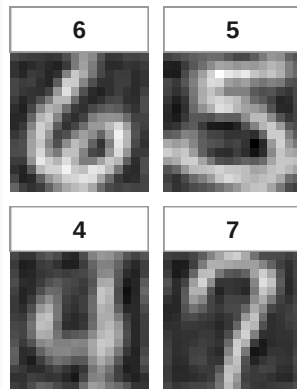


PCA and Dimension Reduction: Example (III)

Using the selection rule leads to 44 components. Using the projection

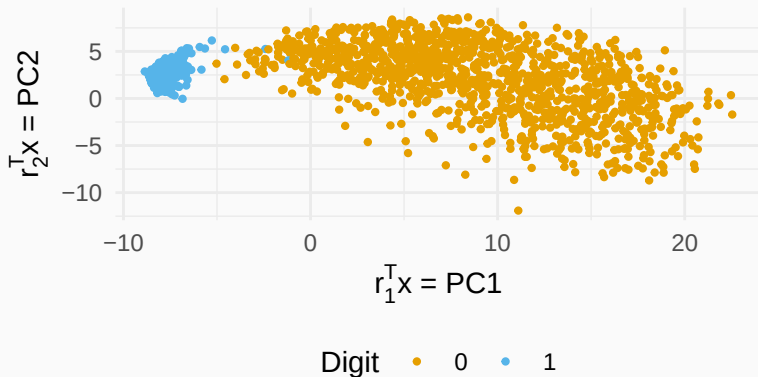
$$\hat{\mathbf{x}} = \left(\sum_{j=1}^{44} \mathbf{r}_j \mathbf{r}_j^T \right) \mathbf{x}$$

creates a **reconstruction** of $\mathbf{x}$.



PCA and Dimension Reduction: Example (IV)

Projecting the digits onto the first two principal component directions gives a very clear distinction of digits 0 and 1.



Importance of standardisation (I)

The overall issue: **Subjectivity vs Objectivity**

(Co-)variance is scale dependent: If we have a sample (size n) of variables x and y , then their empirical covariance is

$$s_{xy} = \frac{1}{n-1} \sum_{l=1}^n (x_l - \bar{x})(y_l - \bar{y})$$

If x is scaled by a factor c , i.e. $z = c \cdot x$, then

$$\begin{aligned} s_{zy} &= \frac{1}{n-1} \sum_{l=1}^n (z_l - \bar{z})(y_l - \bar{y}) \\ &= \frac{1}{n-1} \sum_{l=1}^n (c \cdot x_l - c \cdot \bar{x})(y_l - \bar{y}) = c \cdot s_{xy} \end{aligned}$$

Importance of standardisation (II)

(Co-)variance is scale dependent: $s_{zy} = c \cdot s_{xy}$ where $z = c \cdot x$

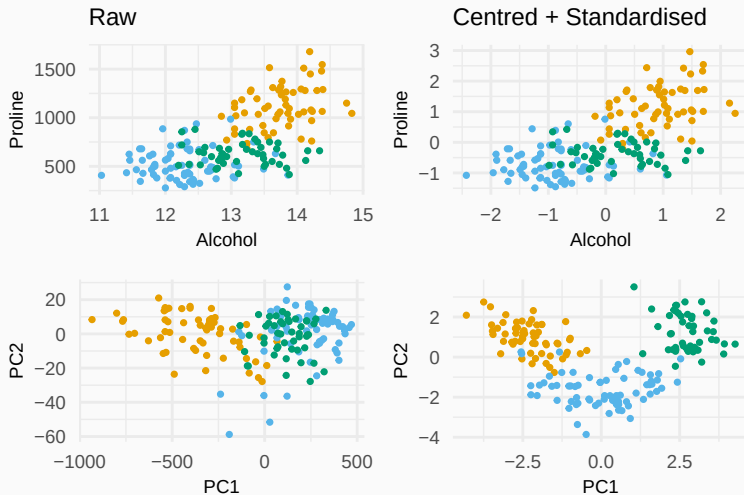
- ▶ By scaling variables we can therefore make them as large/influential or small/insignificant as we want, which is a very **subjective** process
- ▶ By standardising variables we can get of rid of **scaling** and reach an **objective** point-of-view
- ▶ **Do we get rid of information?**
 - ▶ The **typical range** of a variable is compressed
 - ▶ The overall shape of the data is preserved
 - ▶ Outliers will still be outliers

UCI Wine Data Set¹

- ▶ Results of a chemical analysis on multiple samples from three different origins of wine
- ▶ $n = 178$ samples (59 origin 1, 71 origin 2, 48 origin 3)
- ▶ $p = 13$ features
 - ▶ e.g. alcohol in %, ash, colour intensity, magnesium, ...

¹<https://archive.ics.uci.edu/ml/datasets/Wine>

Importance of standardisation (III)



Singular Value Decomposition

Singular Value Decomposition (SVD)

The **singular value decomposition (SVD)** of a matrix $\mathbf{X} \in \mathbb{R}^{n \times p}$, $n \geq p$, is

$$\mathbf{X} = \mathbf{U}\mathbf{D}\mathbf{V}^T$$

where $\mathbf{U} \in \mathbb{R}^{n \times p}$ and $\mathbf{V} \in \mathbb{R}^{p \times p}$ with

$$\mathbf{U}^T \mathbf{U} = \mathbf{I}_p \quad \text{and} \quad \mathbf{V}^T \mathbf{V} = \mathbf{V}\mathbf{V}^T = \mathbf{I}_p$$

and $\mathbf{D} \in \mathbb{R}^{p \times p}$ is diagonal. Usually

$$d_{11} \geq d_{22} \geq \dots \geq d_{pp}$$

Note: Due to the **orthogonality conditions** on $\mathbf{U}$ and $\mathbf{V}$

$$\mathbf{X}\mathbf{X}^T \mathbf{U} = \mathbf{U}\mathbf{D}^2$$

$$\mathbf{X}^T \mathbf{X}\mathbf{V} = \mathbf{V}\mathbf{D}^2$$

In PCA the empirical covariance matrix $\hat{\Sigma}$ is in focus, whereas SVD focuses on the data matrix $\mathbf{X}$ directly.

Connection: For centred variables

$$\hat{\Sigma} = \frac{\mathbf{X}^T \mathbf{X}}{n-1} = \frac{\mathbf{V} \mathbf{D} \mathbf{U}^T \mathbf{U} \mathbf{D} \mathbf{V}^T}{n-1} = \mathbf{V} \left(\frac{\mathbf{D}^2}{n-1} \right) \mathbf{V}^T$$

The PC directions are in $\mathbf{V}$ and the eigenvalues of $\hat{\Sigma}$ are $d_{jj}^2/(n-1)$.

Note: This is how PCA is typically calculated. SVD is a **more general tool** and is used in many other contexts as well.

SVD and best rank- q -approximation / dimension reduction

Write $\mathbf{u}_j$ and $\mathbf{v}_j$ for the columns of $\mathbf{U}$ and $\mathbf{V}$, respectively. Then

$$\mathbf{X} = \mathbf{U}\mathbf{D}\mathbf{V}^\top = \sum_{j=1}^p d_{jj} \underbrace{\mathbf{u}_j \mathbf{v}_j^\top}_{\text{rank-1-matrix}}$$

Best rank- q -approximation: For $q < p$

$$\mathbf{X}_{\textcolor{red}{q}} = \sum_{j=1}^{\textcolor{red}{q}} d_{jj} \mathbf{u}_j \mathbf{v}_j^\top$$

with **approximation error**

$$\|\mathbf{X} - \mathbf{X}_q\|_F^2 = \left\| \sum_{j=q+1}^p d_{jj} \mathbf{u}_j \mathbf{v}_j^\top \right\|_F^2 = \sum_{j=q+1}^p d_j^2$$

Cautionary Remarks

- ▶ PCA with leading eigenvectors may not preserve the signal in your data about the outcome Y - alternatives like PLS (partial least squares), CCA (canonical correlation analysis), etc.
- ▶ Pre-processing (centring and standardisation) is important if data is collected on different scales...
- ▶ but.... Is the relative scale of features a nuisance or informative?
- ▶ What about binary or categorical data?
 - ▶ Binary - may be misleading - careful about centering as well
 - ▶ Categorical/Ordinal - use one-shot encoding
 - ▶ Generalized PCA (MSE loss replaced with Deviance for the distribution family) or MCA (multiple correspondence analysis) - looking for co-occurrences of factor levels.
 - ▶ Lot's of packages - even for mixed type data.

Connections to Discriminant Analysis

Discriminant Analysis and the Inverse Covariance Matrix

From PCA or SVD we get $\hat{\Sigma} = \mathbf{V}\mathbf{D}\mathbf{V}^\top$ where $\mathbf{V}^\top\mathbf{V} = \mathbf{V}\mathbf{V}^\top = \mathbf{I}_p$ and $d_{11} \geq \dots \geq d_{pp} \geq 0$. Then

$$\hat{\Sigma}^{-1} = \mathbf{V}\mathbf{D}^{-1}\mathbf{V}^\top = \mathbf{V}\mathbf{D}^{-1/2}\mathbf{D}^{-1/2}\mathbf{V}^\top = (\hat{\Sigma}^{-1/2})^\top \hat{\Sigma}^{-1/2}$$

where $(\mathbf{D}^{-1/2})_{jj} := 1/\sqrt{d_{jj}}$ and $\hat{\Sigma}^{-1/2} := \mathbf{D}^{-1/2}\mathbf{V}^\top$.

In LDA the term involving the inverse covariance matrix is then

$$\begin{aligned}(\mathbf{x} - \hat{\boldsymbol{\mu}})^\top \hat{\Sigma}^{-1} (\mathbf{x} - \hat{\boldsymbol{\mu}}) &= (\mathbf{x} - \hat{\boldsymbol{\mu}})^\top (\hat{\Sigma}^{-1/2})^\top \hat{\Sigma}^{-1/2} (\mathbf{x} - \hat{\boldsymbol{\mu}}) \\&= (\mathbf{V}^\top (\mathbf{x} - \hat{\boldsymbol{\mu}}))^\top \mathbf{D}^{-1} (\mathbf{V}^\top (\mathbf{x} - \hat{\boldsymbol{\mu}})) \\&= \sum_{j=1} \frac{1}{d_{jj}} (\tilde{x}_j - \tilde{\mu}_j)^2\end{aligned}$$

Inverse of the eigenvalues can lead to **numerical instability**.

Regularised Discriminant Analysis (RDA)

The empirical covariance matrix used by LDA can be **stabilized**:

$$\hat{\Sigma}_{\lambda} := \hat{\Sigma} + \lambda \mathbf{I}_p = \mathbf{V}(\mathbf{D} + \lambda \mathbf{I}_p)\mathbf{V}^{\top}$$

where $\lambda > 0$ is a tuning parameter.

- ▶ Using $\hat{\Sigma}_{\lambda}$ in LDA is called **regularised discriminant analysis (RDA)**.
- ▶ Instead of $1/d_{jj}$ the scaling factors are now $1/(d_{jj} + \lambda)$.
- ▶ For small d_{jj} this can lead to **numerical stability**, whereas large d_{jj} are not much affected.
- ▶ For increasingly large λ the d_{jj} will have diminishing impact and RDA starts to become **nearest centroids**.
- ▶ RDA can be used with QDA as well by considering:

$$\hat{\Sigma}_{i,\lambda} := \underbrace{\hat{\Sigma}_i}_{\text{QDA}} + \lambda \underbrace{\hat{\Sigma}}_{\text{LDA}}$$

Take-home message

- ▶ Principal component analysis gives a convenient decomposition of the variance of the data
- ▶ Pre-processing (centring and standardisation) is important if data is collected on different scales
- ▶ Singular value decomposition is a universal workhorse for in numerical methods